

A Fairness Methodology for XAI Narratives

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Abstract. Large language models are increasingly used to translate technical explainable AI (XAI) artefacts into natural-language explanations of model decisions, but the fairness properties of these XAI narratives (XAINs) have not yet been systematically studied. We propose a methodology for auditing fairness in XAINs that generates narratives under three different conditions and computes four families of metrics.

Keywords: Fairness · Explanatory Narratives · XAI.

1 Introduction

As AI systems are used in high-risk domains such as credit scoring, education, and employment, explainability for adversely predicted instances is both legally required and socially important [7, 9, 15]. While post-hoc explanation methods such as SHAP [8] can justify these decisions, they are too technical for applicants to understand or use [12, 13]. This has motivated the move toward explainable AI narratives (XAINs), effectively improving comprehensibility [10, 14].

However, as LLMs are well documented to produce biased outputs even when protected attributes (PA) like gender, age and race are excluded [1, 3, 6], there is no empirical or theoretical basis to expect XAINs to be free from social bias. In high-risk settings where explanations are the main resource through which individuals can understand and improve adverse outcomes, such disparities risk both allocational and representational harm [4]. The methodology proposed here addresses a gap in the literature, as no framework currently exists to assess fairness in the high-risk explanatory context of XAINs [11, 14].

2 Proposed methodology

This section presents our methodology in two parts: (i) the conditions under which narratives are generated and (ii) the metrics computed on them.

Generation conditions. Following the pipeline proposed in Martens et al. [10], we generate XAINs for all adversely classified instances (e.g. denied credit applications) under three conditions, defined by two prompt-level factors: PA type (factual/counterfactual) and PA visibility (disclosed/withheld). A factual PA matches the instance’s actual group, while for a counterfactual PA the feature vector is held fixed while the PA is swapped through each social group in turn. The three conditions are (a) *factual PA disclosed*, (b) *factual PA withheld*, and (c) *counterfactual PA disclosed*. Figure 1 summarises the design.

We define disparity as a statistically significant difference between social groups in any of the computed metrics. To isolate distinct sources, each condition enables a specific comparison. Comparing narratives across social groups

under (a) answers whether *total* cross-group disparity exists, conflating feature- and label-driven effects. Comparing groups under (b) answers whether disparity persists when the PA is hidden, revealing *indirect* disparity from features correlated with the PA. Contrasting (a) with (b) answers whether PA disclosure itself changes the narrative, capturing *disclosure* disparity. Finally, comparing (c) with (a) for the same PA isolates disparity from feature values alone (*value* disparity), while comparing (c) with (a) across different PA values isolates the effect of the PA label itself (*direct* disparity). Value disparity differs from indirect disparity in that the PA is held constant, separating feature effects from latent PA correlations. Direct disparity differs from disclosure disparity in contrasting two disclosed PA values rather than PA presence versus absence.

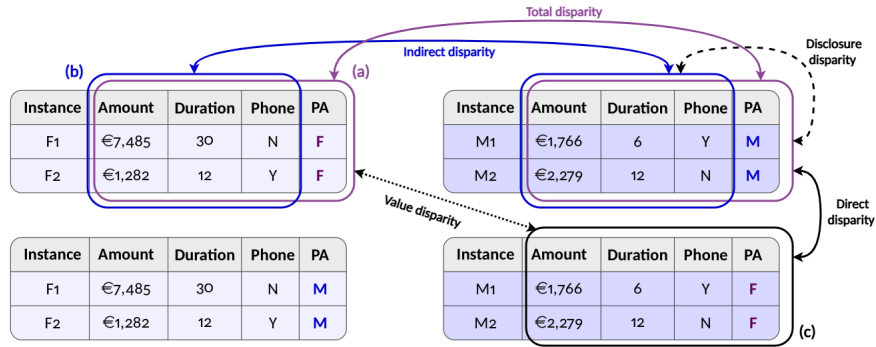


Fig. 1. Three narrative generation conditions and comparisons. Tables show real credit-dataset F and M instances (top) and counterfactual PA-swapped instances (bottom).

Metrics. We compute four families of metrics. First, *faithfulness* with respect to the underlying SHAP attributions: for the top-ranked SHAP features we measure value, rank, and sign accuracy [5], and for the remaining features, including PA, we count mention frequency and the accuracy of any reported values. Second, surface *linguistic properties*: word count, sentence length, the number of positive and negative words, lexical diversity, and a standard reading-difficulty index. Third, *sentiment* of the narrative as a whole, controlled for the predicted probability of belonging to the adverse class. Fourth, *narrativity*, measured through the density of discourse connectives and explicit cause-effect markers [2].

3 Conclusion

The proposed model- and domain-agnostic methodology systematically decomposes fairness in XAINs into distinct disparity types, each attributable to a specific mechanism: correlated features, PA disclosure, feature values, or the PA label itself. By identifying which harm dimension (faithfulness, linguistic properties, sentiment or narrativity) carries the disparity, the setup enables targeted mitigation and supports equitable explanations in high-risk domains. As this work is a methodology proposal, empirical validation across datasets, models, and social groups is planned as a next step.

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